Symmetries in Physics — Exercise Sheet 1

Groups, examples, and Lagrange's theorem

Variant: Questions only

Exercise 1 Let D_n be the dihedral group of order $2n$, generated by a rotation r and a reflection s. Representing $r = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ with $\theta = 2\pi/n$ and $s = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, verify $srs^{-1} = r^{-1}$ directly, and deduce the product rule $(r^a s^\alpha)(r^b s^\beta) = r^{a+(-1)^\alpha b}\, s^{\alpha+\beta}$.

Exercise 2 Show $D_3 \cong S_3$ by an explicit isomorphism between the symmetries of an equilateral triangle and the permutations of its vertices, and verify that it respects one non-commuting product.

Exercise 3 Prove that every group of prime order p is cyclic, hence isomorphic to $\mathbb{Z}/p\mathbb{Z}$ (Lecture 1's $\mathbb{Z}_p$; the quotient notation is explained in Lecture 2).

Exercise 4 The Klein four-group is $V = \{e, a, b, ab\}$ with $a^2 = b^2 = e$ and $ab = ba$. (i) Show $V \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (here $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ is the set of pairs under componentwise addition — the *external direct product*, studied in Lecture 2). (ii) Show that V is the symmetry group of a non-square rectangle and identify the four symmetries geometrically. (iii) Show $V \ncong \mathbb{Z}/4\mathbb{Z}$, although both have order four. (iv) Show that the symmetry group of the planar anisotropic oscillator with potential $W = \frac{1}{2}(k_x x^2 + k_y y^2)$, $k_x \neq k_y$ — meaning the orthogonal maps of the plane that preserve W, and hence the Hamiltonian $H = \frac{1}{2}\left|\vec{p}\right|^2 + W$ — is again V. (Non-orthogonal linear maps can also preserve W; they change the kinetic term and are not symmetries of the system.)

Exercise 5 List all subgroups of the symmetry group D_4 of the square (order 8), give the order and index of each, and verify Lagrange's theorem throughout.

Exercise 6 Determine the point groups of (i) the water molecule H_2O and (ii) ammonia NH_3, and identify each as an abstract group from this sheet.

Exercise 7 Both D_4 and the cyclic group C_8 have order 8. Compute the multiset of element orders in each, and conclude that the two groups are not isomorphic.

Exercise 8 Prove the case $p = 2$ of Cauchy's theorem: every group G of even order contains an element of order 2. *Guidance:* pair each element with its inverse.